$\mathbb{F}_{p^3}$-Frobenius nonclassical curves $y^n = f(x)$ and their geometry

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$\mathbb{F}_q$-Frobenius nonclassical curves

$$\mathcal{X}: \quad F(x,y)=0 \qquad F(x,y) \in \mathbb{F}_q[x,y]$$

Proj., abs. irreducible, algebraic plane **curve** in $\mathbb{P}^2(\overline{\mathbb{F}_q})$ defined over $\mathbb{F}_q$

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$\iff$ for any simple point $P=(a:b:c)\in\mathcal{X}$,

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example: $\qquad F(x,y)=x^{q+1}+y^{q+1}+1$

$\mathbb{F}_q$-Frobenius nonclassical curves: many $\mathbb{F}_q$-rational points

$$\mathcal{X}: \quad F(x,y)=0 \qquad F(x,y)\in\mathbb{F}_q[x,y]$$

- p : characteristic of $\mathbb{F}_q$
- g : genus of $\mathcal{X}$
- $d=\deg(F)$: degree of $\mathcal{X}$
- N_q : number of $\mathbb{F}_q$-rational points of $\mathcal{X}$

$$\text{Stohr-Voloch(1986):} \quad N_q \leq \frac{p^h(2g-2)+(q+2)d}{2}$$

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$h > 0 \iff \mathcal{X}$ is $\mathbb{F}_q$-Frobenius nonclassical

$\longrightarrow$ curves with many rational points

- good AG codes
- ...

$\mathbb{F}_q$-Frobenius nonclassical curves: minimal value set polys

$f(x) \in \mathbb{F}_q[x]$ nonconstant value set : $V_f = \{f(\alpha) \colon \alpha \in \mathbb{F}_q\}$

$$\left\lfloor \frac{q-1}{\deg(f)} \right\rfloor + 1 \leq |V_f| \leq q$$

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Carlitz et al. (1961), Mills (1964), Wan et al. (1993),

Borges (2016), Borges-Conceicao (2018), Borges-Reis (2021, 2025)...

- results on MVSPs in terms of $V_f \subset \mathbb{F}_q$
- $q = p^k$ classification of MVSPs for $k \le 3$
- Borges (2016):

  MVSPs over $\mathbb{F}_q$ $\iff$ certain $\mathbb{F}_q$-Frobenius nonclassical curves

$\mathbb{F}_q$-Frobenius nonclassical curves: minimal value set polys

$f(x), g(x) \in \mathbb{F}_q[x] \setminus \mathbb{F}_q$

$\mathcal{X}: \; g(y) = f(x)$ absolutely irreducible curve

- $\mathcal{X}$ $\mathbb{F}_q$-Frob. nonclassical $\Longrightarrow$ $f(x), g(x)$ are MVSPs, $V_f = V_g$
- $f(x), g(x)$ MVSPs, $V_f = V_g$, $|V_f| > 2$ or $|V_f| = 2 = p$
  $\Longrightarrow$ $\mathcal{X}$ is $\mathbb{F}_q$-Frob. nonclassical

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Borges-Reis (2021): classification of MVSPs over $\mathbb{F}_{p^3}$ $\implies$

$\mathcal{X}: \ y^n = f(x)$ abs. irred., $1 < n < p^3 - 1$, is $\mathbb{F}_{p^3}$-Frob. nonclassical if and only if $n = p^2 + p + 1$ and

$$f(x) = \alpha x^{p^2+p+1} + \beta x^{p^2+p} + \beta^p x^{p^2+1} + \beta^{p^2} x^{p+1} + \gamma^{p^2} x^{p^2} \gamma^p x^p + \gamma x + \delta$$

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$$f_0(x) = x^{p^2+p+1} + 1 \qquad f_1(x) = x^{p^2} + x^p + x \qquad f_2(x) = x^{p^2+1} + x^p$$

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For $\mathcal{X}_i$, $i = 0, 1, 2$, we compute:

- number $|\mathcal{X}_i(\mathbb{F}_{p^3})|$ of $\mathbb{F}_{p^3}$-rational points
- genus $g(\mathcal{X}_i)$
- Automorphism group $\mathrm{Aut}(\mathcal{X}_i) = \{\varphi : \mathcal{X}_i \to \mathcal{X}_i \quad \overline{\mathbb{F}_{p^3}}\text{-birationalities} \}$
- p-rank $\gamma(\mathcal{X}_i)$
- a-number $\mathfrak{a}(\mathcal{X}_i)$

Properties of $\mathcal{X}_0 : \ y^{p^2+p+1} = x^{p^2+p+1} + 1$

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Fermat curve over $\mathbb{F}_{p^3}$ $\implies$ well-studied curve: all previous results

- $|\mathcal{X}_0(\mathbb{F}_{p^3})| = p^5 - p^3 - p^2 + 1$ (old)
- $g(\mathcal{X}_0) = \frac{1}{2}\left(p^4 + 2p^3 - p\right)$ (old)
- $\mathrm{Aut}(\mathcal{X}_0) \cong (C_{p^2+p+1} \times C_{p^2+p+1}) \rtimes \mathrm{Sym}(3)$ (old)
- $\gamma(\mathcal{X}_0) = \frac{1}{8}\left(p^4 + 2p^3 + 3p^2 + 2p\right)$ (Borges-Goncalves 2024)
- $\mathfrak{a}(\mathcal{X}_0) = \frac{1}{4}\left(p^4 + 4p^3 + 5p^2 + 2p\right)$ (Montanucci-Speziali 2018)

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- $|\mathcal{X}_1(\mathbb{F}_{p^3})| = p^5 + 1$ (old)
- $g(\mathcal{X}_1) = \frac{1}{2}\left(p^4 + p^3 - p^2 - p\right)$ (old)
- $\mathrm{Aut}(\mathcal{X}_1) \cong (C_p \times C_p) \rtimes C_{p^3-1}$ (Bonini-Montanucci-Zini 2020)
- $\gamma(\mathcal{X}_1) = 0$ (old)
- $\mathfrak{a}(\mathcal{X}_1) = \frac{1}{12}\left(3p^4 + 2p^3 - 3p^2 - 2p\right)$

Properties of $\mathcal{X}_2 : \; y^{p^2+p+1} = x^{p^2+1} + x^p$

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- $|\mathcal{X}_2(\mathbb{F}_{p^3})| = p^5 + p^3 + p^2 + 1$
- $g(\mathcal{X}_2) = \frac{1}{2}\left(p^4 + p\right)$
- $\mathrm{Aut}(\mathcal{X}_2) \cong C_{p^4+p^2+1} \rtimes C_2$
- $\gamma(\mathcal{X}_2) = \frac{1}{8}\left(p^4 + 2p^3 + 3p^2 + 2p\right)$
- $\mathfrak{a}(\mathcal{X}_2) = \frac{1}{12}\left(3p^4 - 2p^3 - 3p^2 + 2p\right)$

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$\lambda \in \mathbb{F}_{q^6} \quad \lambda^{q^4+q^2+1} = 1$ primitive

$\sigma_\lambda : (x, \ y) \mapsto (\lambda^{q^2+q+1} x, \ \lambda^q y) \qquad \tau : (x, \ y) \mapsto (1/x, \ y/x)$

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Tools:

- One-point stabilizers: $\mathrm{Aut}(\mathcal{X})_P \cong$ (Sylow p-subgr) $\rtimes$ (p'-cyclic gr)
- Group actions; structure of the orbits when $|\mathrm{Aut}(\mathcal{X})| > 84(g(\mathcal{X}) - 1)$
- Weierstrass semigroups $H(P)$
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Merci!